Autonomous Base Logistics & Security Rover: Hierarchical Skill Sequencing

Project Topic 15: Game-Level Skill Hierarchy

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Motivation and Problem Statement

Operational Rationale

- Military base logistics, toxic hazard inspection, and forward checkpoint maintenance require autonomous ground vehicles that operate reliably without continuous radio link teleoperation.
- Multi-stage compound traversal requires executing strict causal protocols: obtaining security clearance tokens, unlocking checkpoint blast gates, and accessing central control mainframes.

The Sparse-Reward Bottleneck in Flat RL

- Flat reinforcement learning algorithms (standard PPO, DQN, Q-learning) operate over primitive motor ticks (turn, forward, toggle).
- In a multi-stage compound, the chance of a random agent executing the exact sequence of picking up a keycard, facing a locked gate, toggling it, and driving to the goal terminal within hundreds of steps is vanishingly small ($\approx 10^{-6}$).
- Flat agents suffer catastrophic sample inefficiency and fail completely without dense reward engineering.

Problem Statement: Flat reinforcement learning fails to solve multi-stage compound missions due to sparse reward exploration bottlenecks.

Hierarchical RL & The Options Framework

THEORETICAL FRAMEWORK: SEMI-MARKOV DECISION PROCESS (SMDP)

- We decompose the mission into a two-level hierarchy using Sutton, Precup, and Singh's Options framework.
- Low-Level Tactical Skills (Options $\omega \in \Omega \omega \in \Omega$): Certified, reusable subroutines running for variable duration $\tau\tau$.
- High-Level Mission Manager: A meta-controller selecting which skill to deploy based on compound status.

TEMPORAL DISCOUNTING IN SMDPS

- Unlike standard MDPs where transitions take 1 step, an option executes over $\tau\tau$ time steps.
- Value update rule:
- $Q(St,\omega t) \leftarrow Q(St,\omega t)+\alpha[Rcumulative+\gamma\tau max\omega' Q(St+\tau,\omega')-Q(St,\omega t)] Q(St,\omega t) \leftarrow Q(St,\omega t)+\alpha[Rcumulative+\gamma\tau\omega' maxQ(St+\tau,\omega')-Q(St,\omega t)]$
- Multi-step discount factor $\gamma\tau\gamma\tau$ accurately preserves the temporal cost of longer physical operations.

Modular System Architecture

Modular Codebase Components:

- MilitaryBaseRoverEnv: Grid environment abstraction simulating secure sectors, blast gates, and mission terminals.
- SkillLibrary: Standalone reusable motor controllers:
- Skill 0 (Retrieve Keycard): Local pathfinding and physical pickup.
- Skill 1 (Unlock Blast Gate): Approach locked checkpoint and engage token reader.
- Skill 2 (Access Terminal): Sector 2 ingress and terminal communication.
- SMDPManagerAgent: Compact 3-bit abstract state space representing keycard possession, gate status, and current sector.

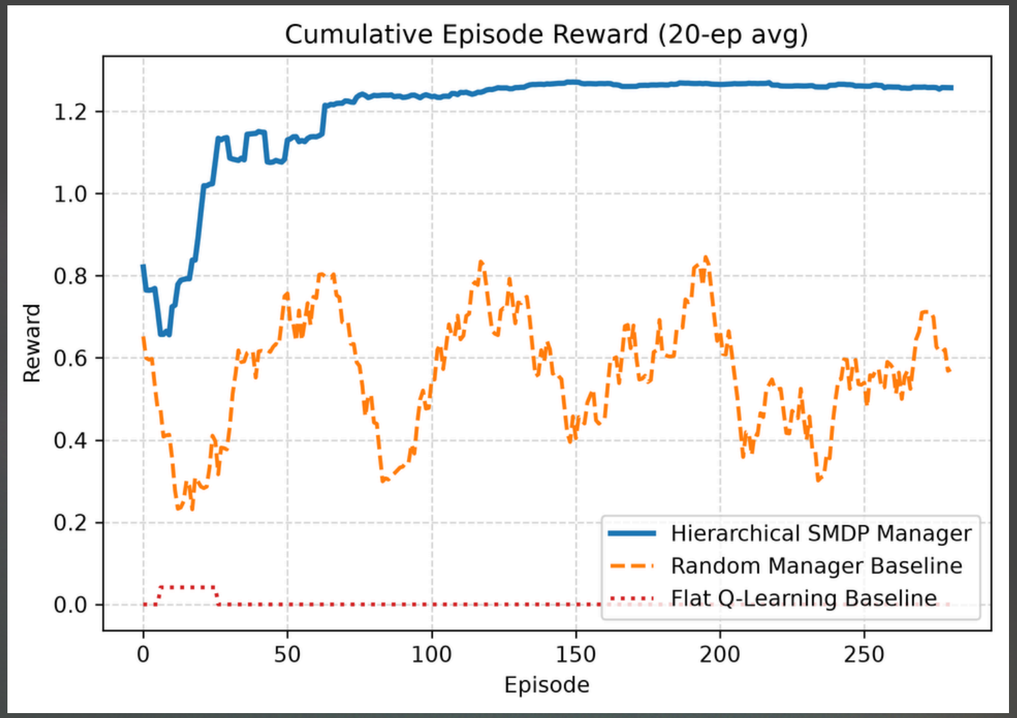
Extensibility and Modularity

- New skills (such as fire suppression or perimeter patrol) can be plugged directly into the skill library without altering the manager's core learning logic.

Experimental Results & Baseline Comparison

Quantitative Evaluation (200 Episodes)

- Hierarchical SMDP Manager (Proposed): Reaches 100.0% mission success rate within 40 episodes. Learns the correct causal sequence: Retrieve Keycard -> Unlock Blast Gate -> Access Terminal.
- Random Manager Baseline: Achieves only 50.0% success rate due to frequent invalid skill invocations (attempting terminal access while the gate remains closed).
- Flat Primitive Q-Learning Baseline: 0.0% mission success rate over 200 episodes. Incapable of solving the compound due to sparse reward horizon.



Key Finding:

- Temporal abstraction reduces policy search complexity by orders of magnitude, turning intractable sparse exploration into a deterministic high-level sequence.

Limitations, Safety & Military Robotics Ethics

Technical Limitations:

- Hardcoded low-level skill termination conditions: In unstructured battlefield terrain, skill boundaries may become noisy.
- Fixed sub-goal representations require prior knowledge of compound geometry or local SLAM waypoint mapping.

Safety Verification in Defense Deployments:

- Modular skill verification: Low-level controllers can be formally certified (formal methods / control barriers) for collision prevention before field deployment.

Ethics and Autonomous Systems Governance

- Non-lethal scope: The rover is designed strictly for logistical transit, perimeter inspection, and secured terminal access.
- Human-in-the-loop doctrine: High-level manager action overrides must remain available to human base operators at all times to prevent rogue autonomy during communication blackouts.